

Leah Song

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TECHNICAL SKILLS

Software: C/C++, Python, MATLAB, ARM, SolidWorks, AutoCAD, Altium, Verilog

Protocols/Systems: IEC 61850 GOOSE, Modbus, IEEE C37.118, DCS

EDUCATION

University of British Columbia

Bachelor of Applied Science in Electrical Engineering (Dean's List Honors 2025-Present)

Vancouver, BC

Sep. 2023 – Present

TECHNICAL EXPERIENCE

Undergraduate Academic Assistant

University of British Columbia

Jan. 2026 – Present

Vancouver, BC

- Supported project management for the Silicon Photonic Biosensor Group and the Mend the Gap Biomaterials Platform Group at the UBC BioMEMS Laboratory through improving workflows and collaborative discussions.
- Efficiently managed lab activities through inventory, documentation, enforcing lab protocols, and working with a diverse interdisciplinary team.

Electrical Engineering Intern

Shell Canada Limited

May 2025 – Dec. 2025

Fort Saskatchewan, AB

- Worked under Electrical Engineering team at Scotford Production Site, managing power systems across refinery, upgraders, chemicals plant, and carbon capture facilities.
- Spearheaded digital fault recording system in 35kV substation using IEEE C37.118 fiber network to record real-time data from HV Siemens relay trips and allow for efficient diagnostics.
- Delivered high-priority operational technology (OT) audit for Shell Global and crucial cybersecurity controls on Rockwell/Siemens/SEL devices to protect digital wireless systems from external threats.
- Overhauled fin fan electric motors from 15 HP to 50 HP, VFDs, and dV/dT filters in central refinery processing unit and delivered 40% higher cooling capacity.

Electrical Lead/Captain

UBC ThunderBots

Jan. 2026 – Present

Vancouver, BC

- Qualified for competitive 15-seat global autonomous robotics roster, successfully placing 2nd at RoboCup Eindhoven 2024.
- Coordinated rebuild of electrical PCB stack-up of motor drivers, power board, and breakbeam assembly to fulfill size constraints while maximizing signal integrity.
- Revamped main 24 V power board, evaluating electrical layout of solenoid auto discharge, voltage regulators, ESP32 MCU, and UI/RasPi connections for manufacturing viability and HV noise control.

Electrical/Mechanical Sub-Team Member

UBC ThunderBots

Sep. 2023 – Dec. 2025

Vancouver, BC

- Created inertial motion unit (IMU) PCB to supply 3D dynamic movement data to firmware in order to refine autonomous behavior and motion planning.
- Designed HV feedback sensing circuit on power board to offer real-time feedback on capacitor charging times and solenoid discharge.

Internal Engagement Member

Engineers Without Borders

Jan. 2024 – Sep. 2024

Vancouver, BC

- Led EWB ACCESS Venture initiatives to construct an affordable, non-invasive cervical cancer/HPV testing kit suited to sub-Saharan African communities
- Communicated with biomedical engineering professors and reproductive health experts to understand areas of cultural sensitivity to be applied in design

PERSONAL PROJECTS

- WellNest Smart Dispenser** | *C, Python* Mar. 2026
- Developed a complete hardware and software prototype for an automated feminine hygiene product dispenser
 - Awarded the Major League Hacking (MLH) Sponsorship Prize at the CMD-F 2026 Hackathon for unique conceptual design and detailed technical execution.
- Autonomous Coin-Picking Robot** | *C, Python* Mar. 2025 – Apr. 2025
- Devised a self-driving robot with an electromagnetic servo assembly to detect and collect metal coins.
 - Programmed firmware to test and strategically plan best motion patterns to optimize performance.
- Reflow Oven Controller** | *ARMv7, Python* Jan. 2025 – Mar. 2025
- Constructed a reflow oven controller system with LCD and thermocouple to effectively manage temperature.
 - Allows user to input PCB soldering settings and graph temperatures through engaging app interface.
- DE1-SoC FPGA** | *SystemVerilog, ARMv7* Sep. 2024 – Dec. 2024
- Implemented SystemVerilog and ARM Assembly to design finite state machine controllers and arithmetic datapaths.
 - Established designs under both combinational and sequential logic, designing projects such as a combinational lock, basic mathematics calculator, and a binary operator.
- Autonomous Claw** | *C++, CAD* Mar. 2024 – May 2024
- Developed an autonomous claw from scratch, including manufacturing of metal sheets, hinges and rubber lining.
 - Integrated Arduino Uno and servo motor to pick up various objects based on sensor recognition.

OTHER WORK EXPERIENCE

- Program Leader / Recreation Instructor** Mar. 2023 – Present
City of Coquitlam *Coquitlam, BC*
- Mediated groups of 20-24 clientele and developed tailored lesson plans catered to their respective mental/physical capabilities
 - Worked in high pressure situations requiring patience and confidentiality, particularly with those diagnosed with special needs
- Specialty Instructor** Jan. 2021 – Jul. 2023
City of Port Moody *Port Moody, BC*
- Developed detailed lesson plans according to HIGH-FIVE policy guidelines, including objectives and management strategies

VOLUNTEER EXPERIENCE

- Hospital Wayfinder** Jun. 2025 - Dec. 2025
Alberta Health Services *Edmonton, AB*
- Greets visitors at University of Alberta Hospital, assists patients to appointments, offers directions to relevant clinics, helps patients with mobility issues.
- EUS Council Secretary** May 2024 - May 2025
UBC Engineering Undergraduate Society *Vancouver, BC*
- Records official meeting minutes at all EUS Council Meetings, detailing important information ranging from governing budgets to faculty-wide student events under professional decorum

ACTIVITIES AND INTERESTS

- Competitive Soccer** (Div. 1)
- Dance** (Royal Academy of Dance Ballet)
- Child Development** (HIGH-FIVE special needs certified, recreational therapy)
- Languages** (English, Korean, French, Spanish, Danish)